Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
9.	(a)	(i)		electrons in an atom are excited and the additional energy promotes them to higher energy levels (1) (when the source of energy is removed) the electrons fall back down to a lower energy level emitting energy/light (in the form of a photon) (1)	2			2		
		(ii)		award (1) for either of following absorption spectrum comprises dark lines while emission spectrum comprises coloured lines absorption spectrum has coloured background while emission spectrum has dark background	1			1		
	(b)			$f = \frac{3.00 \times 10^8}{95 \times 10^{-9}}$ (1) $f = 3.16 \times 10^{15}$ Hz $= 3.16 \times 10^9$ MHz (1)		2		2	2	
	(c)	(i)		exothermic since percentage product decreases as temperature increases (1) system opposes increase in temperature and takes in heat by shifting equilibrium to the left /endothermic direction (1)			2	2		
		(ii)		decrease in moles since percentage product increases as pressure increases(1) system opposes increase in pressure by shifting equilibrium to the right / to the side of fewer moles of gas (1)			2	2		

Question				Marking details	Marks available					
					AO1	**AO2**	**AO3**	**Total**	**Maths**	**Prac**
	(d)			moles NaCl = $\frac{15 \times 10^6}{58.5}$ = 2.56×10^5 (1) moles Na_2CO_3 = 1.28×10^5 mass Na_2CO_3 = $1.28 \times 10^5 \times 106$ = 13 568 000 g (1) mass Na_2CO_3 = 1.36×10^4 kg (1) final answer **must** be given to 3 sig figs		1 1	1	3	3	
				Question 9 total	**3**	**4**	**5**	**12**	**5**	**0**